

Isaac Mubiru

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PROFESSIONAL SUMMARY

Senior backend engineer with 5+ years building APIs that other developers depend on — versioned gRPC contracts across microservices serving 4 NGO tenants, Firebase Cloud Functions APIs at MTailor and Dr Wealth, Web3 contract integrations at Mind2matter. Code at github.com/zac-09.

Led the migration of 20+ apps from Parse/MongoDB to Firebase at MTailor with zero downtime, saving ~\$5K/month. Strong fit for partner-facing API platform work — every project I've shipped has been API-first or had downstream consumers depending on stable contracts. Python fluent on data pipelines and migration scripts; Django is the natural ramp from my Express.js MVC depth. Async-first remote: 5+ years on US-based teams from Uganda (UTC+3).

CORE COMPETENCIES

API Design (gRPC + REST)	Node.js / Express	TypeScript / JavaScript	Python (data + scripting)	OAuth / JWT / Firebase Auth
Third-Party Integrations	Microservices	Apache Kafka / NATS / gRPC	GCP / AWS / Firebase	Docker / Kubernetes
Zero-Downtime Migrations	Tech Lead (4 engineers)	Async Remote		

WORK EXPERIENCE

MTailor Sep 2022 – Present

Software Engineer · US Remote (Uganda, UTC+3)

- Led the end-to-end migration of **20+ production applications** from Parse/MongoDB to Firebase/GCP with **zero downtime**, reporting directly to the CTO.
- Designed and implemented real-time two-way sync between MongoDB and Firestore using Node.js + Google Pub/Sub — parallel write path with drift detection.
- Built migration tooling in Python — read MongoDB, applied transformation logic, wrote Firestore + GCS.
- Wrote internal SDK guides and trained the Ops team on Firebase Dashboard + backup procedures.
- Saved **~\$5,000/month** by migrating off AWS.

CodeBits Jan 2020 – Jul 2021

Software Engineer · Lead of 4 · API Platform for NGOs

- Architected a **microservices backend with versioned gRPC contracts** shared across services and 4 NGO tenants — typed APIs that downstream teams could consume confidently.
- Apache Kafka topics for events, NATS Streaming for ordered streams, gRPC for inter-service sync, Docker + Kubernetes for deploy.
- Built a USSD legal-aid service over flaky 2G — Node.js backend with retry-aware session state machine.
- Built a cross-platform mobile app for LASPNET using React Native + Expo with Firebase push.
- Led code review culture and paired architecture sessions; team retained engineers and shipped 3 products in parallel.

Dr Wealth Aug 2021 – Jan 2022

Software Engineer · Singapore Remote

- Extended customer-facing backend APIs using Firebase Cloud Functions and Express.js on Heroku.
- Built ad-hoc analytical jobs querying **2M+ Firestore records**, keeping prices current from Morningstar APIs.

Mind2matter.co Jan 2022 – Jul 2022

Software Engineer · US Remote

- Built backends for DeFi applications using Web3 + Node.js — third-party contract integrations, ECDSA wallet signing, off-chain indexing.
- Delivered React UIs from Figma designs under tight client timelines.

EDUCATION

B.Sc. Computer Engineering — **Makerere University** Aug 2018 – Sep 2022

SKILLS

Proficient: Node.js, Express, TypeScript, JavaScript, gRPC + Protobuf, REST API design, OAuth/JWT, Firebase Auth, Python (scripting + data), Apache Kafka, NATS Streaming, Firebase/GCP, MongoDB, SQL, Docker, Kubernetes, Linux, Git

Intermediate: Go, Rust, AWS, React, React Native, Web3

Open to ramp: Django + DRF (Express.js MVC mental model translates directly)

Languages: English (native-level), Kiswahili